

|          |                                                                                                                                    |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          |                                                                                                                                    |           |
| 3(a)(i)  | $E = \frac{1}{2}Fx$ <b>or</b> $E = \frac{1}{2}kx^2$ <b>or</b> $E = \text{area under graph}$                                        | <b>C1</b> |
|          | $E = \frac{1}{2} \times 4.0 \times 0.32 = 0.64 \text{ J}$ <b>or</b> $E = \frac{1}{2} \times 12.5 \times (0.32)^2 = 0.64 \text{ J}$ | <b>A1</b> |
| 3(a)(ii) | $E = mgh$ <b>or</b> $E = Wh$<br>$= 2.5 \times 0.32$                                                                                | <b>C1</b> |
|          | $= 0.80 \text{ J}$                                                                                                                 | <b>A1</b> |
| 3(b)(i)  | kinetic energy $= 0.80 - 0.64$<br>$= 0.16 \text{ J}$                                                                               | <b>A1</b> |
| 3(b)(ii) | $E = \frac{1}{2}mv^2$<br>$0.16 = \frac{1}{2} \times (2.5 / 9.81) \times v^2$                                                       | <b>C1</b> |
|          | $v = 1.1 \text{ m s}^{-1}$                                                                                                         | <b>A1</b> |